

ECUACIÓN LINEAL: $ax+b=0$ $a \neq 0$

- Soluciones:
- Solución única $x=c$ $C_S: \{c\}$ $C_S: \{x=c\}$
 - Infinitas soluciones $0 \cdot x = 0$ $C_S: \mathbb{R}$ $C_S: (-\infty, \infty)$
 Ej: $x=3 \Rightarrow 0 \cdot 3 = 0 \checkmark$
 $x=-100 \Rightarrow 0 \cdot -100 = 0 \checkmark$
 - Conjunto solución vacío (sin solución) $0 \cdot x = c \wedge c \neq 0$
 Ej: $c=4$, $x=2 \Rightarrow 0 \cdot 2 = 4 ?$ $C_S: \emptyset$
 $0 = 4 \times \text{falso}$
 $x=-10 \Rightarrow 0 \cdot (-10) = 4 ?$ $C_S: \{\}$
 $0 = 4 \text{ falso}$

Ejercicio 1: Halle el valor de la incógnita y verifique

- a) $3x-1=11$ b) $\frac{13-5x}{3}=1$ c) $3A-1=A+3$ d) $\frac{10}{M}-4=1$
 e) $4C=7C+3$ f) $\frac{2-5N}{6}=N+4$ g) $P+4=P$ h) $3B+1=2B+B+1$

f) $\frac{2-5N}{6} = N+4$

$$2-5N = (N+4) \cdot 6$$

$$2-5N = 6N+24$$

$$-5N-6N = 24-2$$

$$-11N = 22$$

$$N = \frac{22}{-11}$$

$$\boxed{N = -2} \quad C_S: \{-2\}$$

$$\frac{2-5N}{6} \cdot 6 = (N+4) \cdot 6$$

Verificación

$$\frac{2-5N}{6} = N+4$$

$$\frac{2-5 \cdot (-2)}{6} = -2+4$$

$$\frac{2+10}{6} = 2$$

$$\frac{12}{6} = 2$$

$$2 = 2 \checkmark$$

Ejercicio 2: Resuelva y verifique las siguientes ecuaciones:

a) $4(2x-1)+5x=x-2(1-x)-2(x+1)$ b) $\frac{5}{2} + \frac{2-x}{3} - \frac{3+x}{3} = \frac{1}{10}(x-2)$

c) $\frac{x-1}{3} + \frac{2-3x}{4} = -\frac{x}{3}$

(d) $\frac{1}{x-1} - \frac{1}{2+x} = 0$

(e) $\frac{2}{x} - \frac{3}{2x} = \frac{1}{2} \left(\frac{4}{x} - \frac{1}{x} \right)$

f) $3 - \frac{5-2x}{5} = 4 - \frac{4-7x}{10} + \frac{x+2}{2}$

g) $\frac{3}{2}x = x + \frac{x}{2} + 2$

h) $\frac{3}{2}x = x + \frac{x}{2} + 2$

i) $4, 1, 1, 10$

j) $0, 2$

$$g) \frac{x}{2} = x + \frac{x}{2}$$

$$i) \frac{4}{x-2} + \frac{1}{x-3} = \frac{1}{2} + \frac{10}{x}$$

$$h) \frac{x}{2} = x + \frac{x}{2} + 2$$

$$j) \frac{0}{x} = 2$$

$$d) \frac{1}{x-1} - \frac{1}{2+x} = 0$$

$$\frac{(2+x) \cdot 1 - (x-1) \cdot 1}{(x-1)(2+x)} = 0$$

$$(2+x) - (x-1) = 0 \cdot (x-1)(2+x)$$

$$2+x - x + 1 = 0$$

$$x - x = 0 - 2 - 1$$

$$0x = -3 \quad C_S: \emptyset, \quad C_S: \{ \}$$

$$\frac{1}{(x-1)} - \frac{1}{(2+x)} = \frac{(2+x) \cdot 1 - (x-1) \cdot 1}{(x-1)(2+x)}$$

$$\frac{(x-1)(2+x)}{(x-1)}$$

$$x - x = (1-1)x = 0x$$

$$e) \frac{2}{x} - \frac{3}{2x} = \frac{1}{2} \left(\frac{4}{x} - \frac{1}{x} \right)$$

$$\frac{2}{x} - \frac{3}{2x} = \frac{1}{2} \cdot \frac{4}{x} - \frac{1}{2} \cdot \frac{1}{x}$$

$$\frac{2}{x} - \frac{3}{2x} = \frac{2}{x} - \frac{1}{2x}$$

$$\frac{2}{x} - \frac{3}{2x} - \frac{2}{x} + \frac{1}{2x} = 0$$

$$-\frac{3}{2x} + \frac{1}{2x} = 0$$

$$\frac{-3 + 1}{2x} = 0$$

$$-2 = 0 \cdot 2x$$

$$\boxed{-2 = 0}$$

$$-2 = 0 \cdot 2x$$

$$-2 = 0x$$

$$0x = -2$$

$$\boxed{C_S: \emptyset}$$

$$-\frac{3}{2x} + \frac{1}{2x} = \frac{\quad}{2x}$$

Verificación

$$\frac{2}{x} - \frac{3}{2x} = \frac{1}{2} \left(\frac{4}{x} - \frac{1}{x} \right)$$

$$\frac{x=1}{\frac{2}{1} - \frac{3}{2} = \frac{1}{2} \left(\frac{4}{1} - \frac{1}{1} \right)}$$

$$2 - \frac{3}{2} = \frac{1}{2} \cdot 3$$

$$\frac{1}{2} = \frac{3}{2} ? \text{ Falso}$$

$$\boxed{\begin{array}{l} \text{CA} \\ \frac{2-3}{1 \cdot 2} = \frac{4-3}{2} \\ = \frac{1}{2} \end{array}}$$

$$i) \frac{4}{x-2} + \frac{1}{x-3} = \frac{1}{2} + \frac{10}{x}$$

$$\frac{4}{x-2} + \frac{1}{x-3} - \frac{10}{x} - \frac{1}{2} = 0$$

$$\frac{4(x-3) \cdot x \cdot 2 + 1 \cdot (x-2) \cdot x \cdot 2 - 10(x-2)(x-3) \cdot 2 - 1(x-2)(x-3) \cdot x}{(x-2)(x-3) \cdot x \cdot 2} = 0$$

$$4(x-3) \cdot x \cdot 2 + 1 \cdot (x-2) \cdot x \cdot 2 - 10(x-2)(x-3) \cdot 2 - 1(x-2)(x-3) \cdot x = 0 \cdot (x-2)(x-3) \cdot x \cdot 2$$

$$\frac{(x-2)(x-3) \cdot x \cdot 2}{x} = (x-2)(x-3) \cdot 2$$

$$8x^1 \cdot x^1 = 8x^2$$

$$\boxed{\begin{array}{l} \text{CA} \\ 8x(x-3) \\ = 8x^2 - 24x \end{array}}$$

$$4(x-2)(x-3) + 1(x-2)(x-3) - 10(x-2)(x-3) = 0$$

$$(x-2)(x-3) \cdot X \cdot 2$$

$$8x(x-3) + 2x(x-2) - 20(x-2)(x-3) - x(x-2)(x-3) = 0 \cdot (x-2)(x-3) \cdot X \cdot 2$$

$$8x^2 - 24x + 2x^2 - 4x - 20(x^2 - 3x - 2x + 6) - x(x^2 - 3x - 2x + 6) = 0$$

$$8x^2 - 24x + 2x^2 - 4x - 20x^2 + 60x + 40x - 120 - x^3 + 3x^2 + 2x^2 - 6x = 0$$

$$-x^3 + (8+2-20+3+2)x^2 + (-24-4+60+40-6)x - 120 = 0$$

$$-x^3 - 5x^2 + 66x - 120 = 0$$

$$(x-4)(-x^2 - 9x + 30) = 0$$

	x^3	x^2	x^1	x^0
	-1	-5	66	-120
2		-2	-14	104
	-1	-7	52	-16

Ruffini

	x^3	x^2	x^1	x^0
	-1	-5	66	-120
4		-4	-36	120
	-1	-9	30	0

$C_S = \{-\frac{13}{4}\}$

$$f) 3 - \frac{5-2x}{5} = 4 - \frac{4-7x}{10} + \frac{x+2}{2}$$

$$-\frac{5-2x}{5} + \frac{4-7x}{10} - \frac{x+2}{2} = \frac{\quad}{10}$$

$$-\frac{5-2x}{5} + \frac{4-7x}{10} - \frac{x+2}{2} = 4-3$$

$$\frac{-(5-2x) \cdot 2 + (4-7x) \cdot 1 - (x+2) \cdot 5}{10} = 1$$

$$-2(5-2x) + 4-7x - 5(x+2) = 1 \cdot 10$$

$$-10 + 4x + 4 - 7x - 5x - 10 = 10$$

$$4x - 7x - 5x = 10 + 10 - 4 + 10$$

$$-8x = 26$$

$$x = \frac{26}{-8} = -\frac{13}{4}$$

$$x = -\frac{13}{4}$$

Ejercicio 3: Despejar la incógnita indicada, dando condiciones de validez

i. $V = \pi R^2 h - \pi r^2 h$ "h"

ii. $S = \frac{a-r}{l-r}$ "r"

• Denominadores
• Raíces

iii. $s = \frac{x-ar}{w+2rw}$; "w"

iv. $M = \frac{w}{F} \left(1 - \frac{g}{q} \right)$ "g"

$$ii) \quad S = \frac{a - rl}{l - r} \quad "r"$$

$$S \cdot (l - r) = a - rl$$

$$sl - sr = a - rl$$

$$-sr + rl = a - sl$$

$$r(-s + l) = a - sl$$

$$r = \frac{a - sl}{-s + l}$$

condiciones
de validez : $\frac{l - r \neq 0}{l \neq r}$

$$\underline{\underline{-s + l \neq 0}}$$